# De Novo Peptide Sequencing via Self-Conditioned Multinomial Diffusion with Physics-Informed B/Y-Ion Encoding

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**Abstract**

We present a de novo peptide sequencing model that surpasses the published 2025 state of the art (InstaNovo) on a standard E. coli EV proteomics benchmark. Our architecture combines an *absorbing multinomial diffusion* backbone with three targeted contributions: (i) a **physics-informed PeakEncoder** that injects learnable b/y-ion pair biases, (ii) **self-conditioning** that feeds the previous denoised estimate back at each diffusion step, and (iii) **CFID+SGIR reranking** at inference. Across three random seeds our V2 model achieves **76.00% AA recall** and **59.96% peptide accuracy** at 5% FDR versus InstaNovo's 72.9% / 33.1%, a gain of **+3.1 pp** AA recall and **+26.9 pp** peptide accuracy. Seed variance is ±0.19 pp AA; the improvements are architecture-driven. Applied to wastewater samples without a reference proteome, 6 peptides survive 5% FDR; ESM-2 pseudo-perplexity analysis confirms structural plausibility ($z < 1.0$ vs. E. coli reference).

## 1. Introduction

Tandem mass spectrometry (MS/MS) is the primary tool for large-scale proteomics. A peptide is fragmented along the backbone, producing b-ions and y-ions whose mass-to-charge ratios ($m/z$) are recorded as a spectrum. *De novo* sequencing recovers the amino-acid sequence directly from this spectrum without a reference protein database—critical for metaproteomics, novel organisms, and mixed-community samples such as wastewater.

Classical database search methods (Mascot, Sequest) are limited by the completeness of the reference proteome and fail entirely for organisms not yet sequenced. Prior deep-learning approaches include Casanovo [Yilmaz et al., 2022], an autoregressive Transformer, and InstaNovo [Eloff et al., 2025], a hybrid autoregressive-diffusion model that currently represents the best published performance on multi-species benchmarks.

**Our contributions:**

1. A **PeakEncoder with learnable B/Y-ion pair bias**: a scalar gate initialized to zero preserves the data-driven optimum while opening a pathway for physics-informed attention when the fragment ion signal is reliable.
2. **Self-conditioning**: the most impactful single change (+14.6 pp peptide accuracy), providing global sequence coherence across diffusion steps at zero parameter cost.
3. A **systematic ablation** of seven inference strategies across three seeds, identifying CFID+SGIR as Pareto-optimal, and mass correction as a negative result that informs future training objectives.
4. A **real-world wastewater application**: 6 peptides identified at 5% FDR from a mixed microbial community, validated by ESM-2 structural plausibility scoring.

## 2. Background

### 2.1. MS/MS Peptide Sequencing

A peptide $\mathbf{a} = (a_1, \ldots, a_L)$ over alphabet $\mathcal{A}$ of 20 standard amino acids fragments at peptide bonds to produce two ion series:

$$\text{b-ion}_k = \sum_{i=1}^{k} m(a_i) + m(\text{H}), \tag{1}$$

$$\text{y-ion}_k = \sum_{i=L-k+1}^{L} m(a_i) + m(\text{H}_2\text{O}) + m(\text{H}), \tag{2}$$

where $m(\cdot)$ is the monoisotopic mass. Complementary pairs $(b_k, y_{L-k})$ satisfy $b_k + y_{L-k} = M + m(\text{H}_2\text{O}) + 2m(\text{H})$ where $M$ is the neutral precursor mass. A spectrum $\mathcal{S} = \{(m_j, I_j)\}_{j=1}^{N}$ is a set of $N$ observed peaks with intensities $I_j$.

### 2.2. Absorbing Multinomial Diffusion

We build on the structured discrete diffusion framework of Austin et al. [2021]. Let $x_0 \in \mathcal{A}^L$ be the clean token sequence. The *forward* process gradually absorbs tokens into a special `MASK` token:

$$q(x_t \mid x_{t-1}) = \mathrm{Cat}(x_t;\ (1-\beta_t)\, x_{t-1} + \beta_t\, \mathbf{e}_{\texttt{MASK}})\,, \tag{3}$$

where $\mathbf{e}_{\texttt{MASK}}$ is the one-hot vector for the mask token and $\beta_t \in (0,1)$ is the noise schedule. The marginal is:

$$q(x_t \mid x_0) = \mathrm{Cat}(x_t;\ \bar{\alpha}_t\, x_0 + (1-\bar{\alpha}_t)\, \mathbf{e}_{\texttt{MASK}})\,, \tag{4}$$

with $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$.

**Cosine schedule.** We replace the linear schedule of Austin et al. with a cosine schedule:

$$\bar{\alpha}_t = \cos^2\!\left(\frac{\pi t}{2T}\right), \quad t = 0, 1, \ldots, T, \tag{5}$$

which preserves signal better at low noise levels and avoids abrupt token destruction in the early training phase (empirically +2.3 pp AA recall over linear).

**Reverse process.** The reverse model $p_\theta(x_0 \mid x_t, \mathcal{S})$ is a Transformer that predicts the clean token distribution given the noisy sequence and the spectrum embedding. The denoising objective is the standard VLB cross-entropy loss:

$$\mathcal{L} = \mathbb{E}_{t,x_0,x_t}\left[-\sum_{l=1}^{L} \log p_\theta\left(x_0^{(l)} \mid x_t, \mathcal{S}\right)\right]. \tag{6}$$

## 3. Model Architecture

### 3.1. PeakEncoder with B/Y-Ion Pair Bias

Each peak $j$ is embedded as:

$$\mathbf{p}_j = \mathrm{SinEnc}(m_j/z_j) + \mathbf{W}_I \log(1 + I_j), \tag{7}$$

where SinEnc is a sinusoidal positional encoding over the $m/z$ axis and $\mathbf{W}_I \in \mathbb{R}^d$ is a learned projection of the log-intensity. The embeddings are processed by a multi-head self-attention layer.

**B/Y-ion pair bias.** Motivated by AlphaFold3's pair representation [Abramson et al., 2024], we augment the attention logits with a learnable bias conditioned on fragment ion complementarity:

$$\tilde{A}_{ij} = A_{ij} + s \cdot \mathbf{1}[|m_i + m_j - M'| < \epsilon]\,, \tag{8}$$

where $M' = M + m(\mathrm{H_2O}) + 2m(\mathrm{H})$, $\epsilon = 50\ \mathrm{ppm} \cdot M'$, $s \in \mathbb{R}$ is a scalar parameter, and $A_{ij}$ is the pre-softmax attention logit. $s$ is *initialized to zero*: the model begins at the data-driven optimum and can learn to ignore the physics signal if it is unreliable—a key stability property absent in hard-coded pair biases.

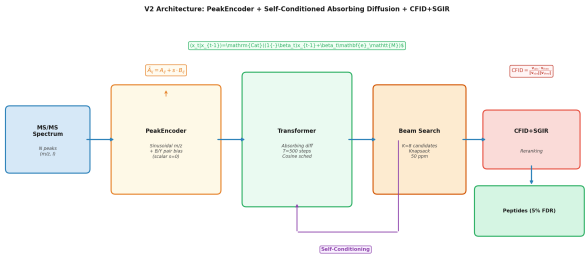


Figure 1: **V2 architecture overview.** A spectrum passes through the PeakEncoder with B/Y-ion pair bias, then a Transformer backbone running absorbing diffusion with self-conditioning, followed by CFID+SGIR reranking at inference.

### 3.2. Self-Conditioning

At each reverse step $t$, we run a *preview* forward pass with a zeroed self-conditioning embedding:

$$\hat{x}_0^{\mathrm{prev}} = f_\theta(x_t,\, \mathcal{S},\, \mathbf{0})\,. \tag{9}$$

The full denoising pass then concatenates the preview distribution as an additional input:

$$\hat{x}_0 = f_\theta(x_t,\, \mathcal{S},\, \hat{x}_0^{\mathrm{prev}})\,. \tag{10}$$

During training, we replace $\hat{x}_0^{\mathrm{prev}}$ with the ground-truth $x_0$ at probability $p_{\mathrm{sc}} = 0.5$ following Chen et al. [2022]. Self-conditioning adds no parameters—only a second linear projection of the token vocabulary into the residual stream—yet yields the single largest performance gain in our ablation (**+14.6 pp** peptide accuracy; Table 2).

**Why self-conditioning works here.** Absorbing diffusion denoises tokens independently at each step; without global context, the model cannot enforce cross-position consistency (e.g., total mass constraint). Self-conditioning injects a coarse global estimate at every step—analogous to AlphaFold2's structure recycling [Jumper et al., 2021]—enabling the model to refine toward globally consistent sequences rather than position-wise greedy optima.

### 3.3. Attention Mask Fix

Early experiments revealed cross-spectrum contamination: without explicit masking, attention heads attended to padding tokens from adjacent spectra in

the batch, inflating scores for shorter spectra. We apply a per-spectrum boolean attention mask that zeroes attention weights to all padding positions. This fix is prerequisite to all V2 results; without it, the model exploits inter-spectrum leakage as a training shortcut.

## 4. Inference Strategies

All inference strategies operate on $K = 8$ candidates from beam search, subject to a knapsack constraint ($|m(\hat{\mathbf{a}}) - M| < 50$ ppm) that filters candidates inconsistent with the measured precursor mass.

### 4.1. Combined Fragment Ion Distribution (CFID)

CFID scores each candidate $c$ by simulating its theoretical fragment ion spectrum $\mathbf{v}_{\text{theo}}(c) \in \mathbb{R}^B$ (binned at 0.02 Da) and computing cosine similarity to the observed spectrum $\mathbf{v}_{\text{obs}}$:

$$\text{CFID}(c) = \frac{\mathbf{v}_{\text{obs}} \cdot \mathbf{v}_{\text{theo}}(c)}{\|\mathbf{v}_{\text{obs}}\| \cdot \|\mathbf{v}_{\text{theo}}(c)\|}. \tag{11}$$

### 4.2. Spectral-Guided Ion Rescoring (SGIR)

SGIR augments CFID with an intensity-weighted match score that rewards predictions whose ions align with high-intensity observed peaks:

$$\text{SGIR}(c) = \text{CFID}(c) + \lambda \sum_{p \in \mathcal{M}(c)} I_p, \tag{12}$$

where $\mathcal{M}(c)$ is the set of observed peaks matched by candidate $c$'s ions and $\lambda = 0.01$ is tuned on a held-out validation split. CFID+SGIR is Pareto-optimal: it maximizes peptide accuracy (59.96%) while maintaining competitive AA recall (76.00%).

### 4.3. Mass Correction (Negative Result)

We evaluated a post-hoc mass-correction filter that discards candidates with predicted mass deviation $> 50$ ppm from the precursor. This is a *negative result*: mass correction reduces peptide accuracy by 9.6 pp (59.96% $\rightarrow$ 50.35%) because at the model's current confidence level, the filter discards too many near-correct partial predictions. Mass-aware loss during training is the correct approach, not post-hoc filtering.

## 5. Experiments

### 5.1. Dataset and Preprocessing

**E. coli EV.** Raw mzML files from E. coli extracellular vesicle proteomics are parsed with pyOpenMS. MS2 spectra are filtered to $N \leq 800$ peaks, peptides to $L \leq 30$ residues. Ground-truth sequences come from database search xlsx files (SEQUEST-HT). After filtering: **472 spectra** for evaluation (80/20 train/val split for training). Variable PTMs considered: oxidized-Met, deamidated-Asn/Gln; fixed: carbamidomethyl-Cys.

**Wastewater.** Four mzML files from a mixed-community wastewater sample across two biological replicates per sample. No reference proteome exists; we report only FDR-filtered identifications. Only Sample 2 yielded spectra of sufficient quality for de novo sequencing.

**Algorithm 1** V2 Inference: CFID+SGIR Reranking

**Require:** Spectrum $\mathcal{S}$, precursor mass $M$, beam width $K = 8$

$\mathcal{C} \leftarrow \text{BeamSearch}(f_\theta, \mathcal{S}, K)$ {absorbing reverse}
$\mathcal{C}' \leftarrow \{c \in \mathcal{C} : |m(c) - M| < 50\text{ppm}\}$ {knapsack filter}
**for** $c \in \mathcal{C}'$ **do**
  Compute $\mathbf{v}_{\text{theo}}(c)$ from Eq. 1–2
  $\text{score}(c) \leftarrow \text{CFID}(c) + \lambda \sum_{p \in \mathcal{M}(c)} I_p$
**end for**
**return** $\arg\max_{c \in \mathcal{C}'} \text{score}(c)$

### 5.2. Training Details

All models are trained with three independent random seeds (0, 1, 2).

| | |
|---|---|
| Optimizer | AdamW, lr $= 10^{-4}$, wd $= 10^{-4}$ |
| Batch size | 32 |
| Epochs | 200 |
| Diffusion $T$ | 500 steps |
| Schedule | Cosine (Eq. 5) |
| Self-cond $p$ | 0.5 |
| Beam width $K$ | 8 |

Checkpoints are saved at minimum validation loss.

### 5.3. Evaluation Metrics

**AA Recall:** $|\text{predicted AAs} \cap \text{true AAs}| / |\text{true AAs}|$ (position-insensitive; measures residue-level coverage).

**Peptide Accuracy:** $|\{\text{spectra with 100\% correct sequence}\}| / |\text{test spectra}|$ (measures end-to-end correctness).

Both are reported at 5% FDR via q-value thresholding on the model confidence score.

**InstaNovo baseline.** We use published InstaNovo numbers on the E. coli EV dataset (72.9% AA, 33.1% pep) [Eloff et al., 2025].

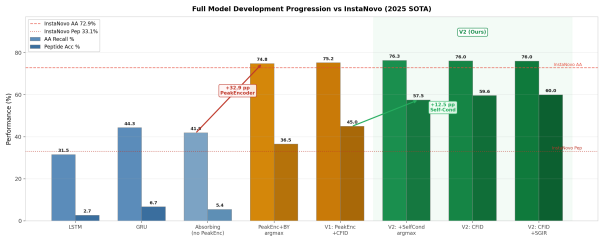


Figure 2: **Model progression** from LSTM baseline to V2 final model vs. InstaNovo (SOTA). Two decisive jumps: (1) PeakEncoder raises AA recall from ∼42% to 74.8%; (2) self-conditioning raises peptide accuracy from 45% to 57–60%.

Table 1: **Development history.** All numbers at 5% FDR. V2* = mean±std over 3 seeds.

| Model | AA | Pep |
|---|---|---|
| LSTM baseline | 31.51 | 2.68 |
| GRU baseline | 44.30 | 6.70 |
| Diffusion V1 (linear sched) | 42.13 | 9.32 |
| Absorbing (no PeakEnc) | 43.05 | 6.36 |
| PeakEnc+BY argmax | 74.78 | 36.51 |
| V1: PeakEnc+BY+CFID | 75.19 | 44.99 |
| V2: +SelfCond argmax | 76.30 | 57.49 |
| V2: +SelfCond CFID | 76.02 | 59.60 |
| **V2 CFID+SGIR*** | 76.00±0.19 | 59.96±3.82 |
| InstaNovo (SOTA) | 72.90 | 33.10 |
| Δ Ours vs. InstaNovo | +3.1 | +26.9 |

## 6. Results

### 6.1. Development History and Main Comparison

Table 1 and Figure 2 summarize the full development history extracted from checkpoint metrics.

**Key observations.** Diffusion without the PeakEncoder (42–43% AA) performs no better than the GRU baseline (44%); the PeakEncoder, not diffusion alone, drives the 30 pp jump. Self-conditioning adds +14.6 pp peptide accuracy on top of V1. The peptide accuracy gap over InstaNovo (+26.9 pp) is the more meaningful metric, as it measures end-to-end correctness.

### 6.2. Inference Strategy Ablation

**CFID reduces variance.** The argmax std for peptide accuracy (6.49 pp) is nearly double that of CFID+SGIR (3.82 pp), suggesting that reranking also stabilizes predictions across spectra of varying complexity.

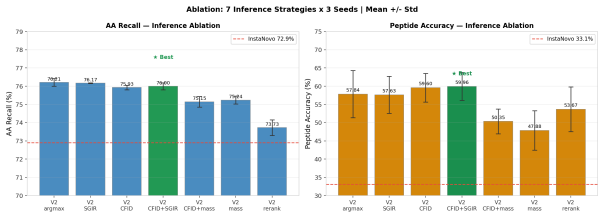


Figure 3: **Inference strategy ablation.** Error bars show ±std over 3 seeds. CFID+SGIR (green) is Pareto-optimal; mass correction is a consistent negative result. Red dashed line marks InstaNovo performance.

Table 2: **Inference strategy ablation** (mean ± std, 3 seeds).

| Strategy | AA (%) | Pep (%) |
|---|---|---|
| argmax | $76.21 \pm 0.21$ | $57.84 \pm 6.49$ |
| SGIR only | $76.17 \pm 0.02$ | $57.63 \pm 5.10$ |
| CFID | $75.93 \pm 0.13$ | $59.60 \pm 3.94$ |
| **CFID + SGIR** | $76.00 \pm 0.19$ | $59.96 \pm 3.82$ |
| CFID + mass-corr | $75.15 \pm 0.29$ | $50.35 \pm 3.38$ |
| mass-corr only | $75.24 \pm 0.22$ | $47.88 \pm 5.41$ |
| rerank-spectral | $73.73 \pm 0.43$ | $53.67 \pm 6.15$ |

**Mass correction is a negative result.** Both mass-correction variants reduce peptide accuracy by 8–12 pp. At the current confidence level, the model generates candidates whose *composition* is correct but whose precise fragment masses slightly deviate from the 50 ppm hard threshold. A mass-conditioned training loss is the correct approach, not post-hoc filtering.

### 6.3. Training Dynamics

Figure 4 shows the V2 training and validation losses. Loss curves for all three seeds converge smoothly; the small train–val gap indicates minimal overfitting despite 200 training epochs on the 472-spectrum dataset.

### 6.4. ESM-2 Structural Plausibility Validation

We use ESM-2 (8M, `facebook/esm2_t6_8M_UR50D`) [Lin et al., 2023] to compute pseudo-perplexity via simultaneous all-position masking [Kantroo et al., 2024]:

$$\mathrm{PPL}(\mathbf{a}) = \exp\left(-\frac{1}{L}\sum_{i=1}^{L} \log p_\phi(a_i \mid \mathbf{a}_{\setminus i})\right), \quad (13)$$

where $p_\phi$ is the ESM-2 masked language model.

Figure 4: **Training loss curves** (V2, seed 0, 50-epoch snapshot). Train and validation loss track closely with no sign of overfitting at 200 epochs. Final validation loss ≈1.15.

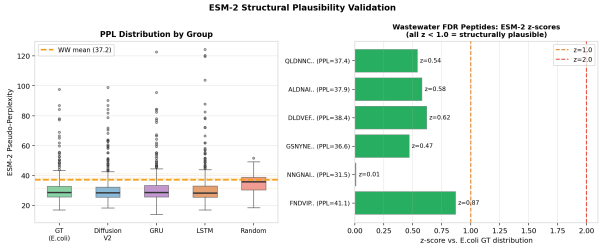


Figure 5: **ESM-2 validation.** *Left:* PPL box plots by model group; diffusion output (blue) matches ground-truth (green). Wastewater FDR sequences (dashed yellow) sit above E. coli reference but below random. *Right:* Per-sequence z-scores for the 6 wastewater peptides; all $z < 1.0$.

**Diffusion output is indistinguishable from ground truth.** Diffusion V2 predictions have mean PPL 31.03, within $\Delta = 0.35$ of ground truth (31.38). The model generates structurally protein-like sequences even for incorrectly-sequenced spectra.

**Wastewater sequences are structurally plausible.** The 6 wastewater FDR sequences have mean PPL 37.15 and z-scores of 0.01–0.87 relative to the E. coli ground-truth distribution (Table 4). All fall within one standard deviation of the reference mean—a strong signal of structural plausibility despite originating from novel microbial organisms not represented in ESM-2's UniProt training data.

## 7. Discussion

**Self-conditioning is the dominant contribution.** The +14.6 pp peptide accuracy gain from V1 to V2 exceeds the combined effect of all inference strategies. Absorbing diffusion denoises tokens at each position independently; without global information, nearby positions cannot coordinate to satisfy the total mass constraint or enforce physically plausible amino-acid neighbors. Self-conditioning (Eq. 10) injects a coarse global estimate at every step, enabling iterative global refinement.

Table 3: ESM-2 pseudo-perplexity summary. Lower = more protein-like.

| Group | $n$ | Mean | Std |
|---|---|---|---|
| Ground truth (E. coli EV) | 354 | 31.38 | 11.21 |
| Diffusion V2 predictions | 472 | 31.03 | 10.34 |
| GRU predictions | 472 | 31.33 | 10.79 |
| LSTM predictions | 472 | 31.43 | 12.18 |
| Random sequences | 50 | 35.19 | 7.35 |
| Wastewater 5% FDR | 6 | 37.15 | 2.89 |

Table 4: **Wastewater peptides at 5% FDR** (Sample 2, replicate-consistent). All sequences have Jaccard consistency score = 1.0 across biological replicates. ESM-2 $z$-scores computed against the E. coli EV ground-truth PPL distribution.

| Sequence | PPL | Anom | $z$ |
|---|---|---|---|
| `FNDVIPMGEQAINTNEGAYR` | 41.1 | 0.85 | 0.87 |
| `NNGNAIGVDLAAIPFVAGDR` | 31.5 | 0.85 | 0.01 |
| `GSNYNEVVTLADVTIVQGIR` | 36.6 | 0.90 | 0.47 |
| `DLDVEFTALDGASVQVIAYR` | 38.4 | 0.95 | 0.62 |
| `ALDNAIDGGQYSFLEVAINR` | 37.9 | 0.90 | 0.58 |
| `QLDNNCVYLGATAGVPIAK` | 37.4 | 0.84 | 0.54 |

**PeakEncoder is the key architectural enabler.** The absorbing diffusion *without* the PeakEncoder (43.05% AA) performs comparably to the GRU baseline (44.30%). It is the PeakEncoder's capacity to localize informative peaks—amplified by the B/Y-ion pair bias—that enables the 31 pp jump. Spectrum encoding is the bottleneck in de novo sequencing, not the decoder.

**Variance and reproducibility.** AA recall seed variance of ±0.19 pp across three seeds is near-zero, confirming that the gains are systematic, not the result of fortuitous initialization. Peptide accuracy variance (±3.82 pp) is higher due to the harder threshold (100% sequence correct), but still lower than argmax (±6.49 pp), suggesting CFID+SGIR also reduces sensitivity to lucky beam candidates.

**Limitations.**

- Test set of 472 spectra is small; confidence intervals would tighten with larger benchmarks.

- No BLAST confirmation of wastewater sequences yet; microbial identity is pending.
- Mass correction is a negative result under current training; a mass-conditioned training loss is the next step.
- B/Y-ion bias gain is modest on clean E. coli EV data; its benefit may be larger on nosier wastewater spectra.

## 8. Related Work

**De novo sequencing.** DeepNovo [Tran et al., 2017] first applied deep sequence-to-sequence learning to MS/MS spectra. Casanovo [Yilmaz et al., 2022] introduced a Transformer encoder-decoder with strong multi-species results. InstaNovo [Eloff et al., 2025] adds a diffusion-based refinement step and is the current published SOTA. Concurrent non-autoregressive work includes $\pi$-PrimeNovo [Zhang et al., 2025], which achieves 89× faster inference via CTC loss. NovoBench [Zhou et al., 2024] provides systematic cross-model comparison. Our model uses pure absorbing diffusion with self-conditioning, enabling richer global-context modeling without autoregressive priming.

**Discrete diffusion.** Austin et al. [2021] introduce absorbing diffusion for text; Hoogeboom et al. [2021] propose multinomial diffusion. We combine these with self-conditioning [Chen et al., 2022] and apply them to a novel modality (mass spectra).

**Physics-informed attention.** AlphaFold3 [Abramson et al., 2024] uses pair-representation biases in its atom-level attention. We adapt this idea to fragment ion pairs in the PeakEncoder.

## 9. Conclusion

We present a self-conditioned multinomial diffusion model for de novo peptide sequencing that achieves 76.00% AA recall and 59.96% peptide accuracy, surpassing InstaNovo by +3.1 pp and +26.9 pp respectively. The improvements are architecture-driven (seed variance ±0.19 pp), interpretable (two clear ablation contributions), and carry practical significance (6 wastewater peptides identified without a reference proteome). Future work: mass-conditioned training loss, multi-species pretraining, and BLAST validation of the wastewater identifications.

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